

Problem Set 3

Matthew Farah
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Problem 1:

(a) Let G be a group of order 27. Prove that G has an element of order 3.

Solution:

To prove that G has an element of order 3, we need to demonstrate that there exists some element $x \in G$ such that $n = 3$ is the smallest positive integer satisfying $x^n = 1$.

So to that end, let's let $a \in G \setminus \{1\}$ be an arbitrary element in G (besides the identity) (which we can do since there are necessarily 26 elements in G besides the identity), and let $m = \text{ord}(a)$ denote the order of a . By Corollary 7.9, we know that m must be a divisor of 27, since $|G| = 27$.

Thus, since the divisors of 27 are 1, 3, 9, and 27, we need only consider four possible cases for m .

(I) $m = 1$.

I lied, this isn't *really* an actual case, as we will soon discover, but is a point needing addressing. Although 1 is indeed a divisor of 27, the order of a cannot possibly be 1 because we defined a to be any arbitrary element of G besides the identity. To see why this is the case, notice that if m did in fact equal 1, we would have that $a^m = 1$, and so:

$$\begin{array}{ll} a^m = 1 & \text{(By definition of order)} \\ \implies a^1 = 1 & \text{(By assumption that } m = 1\text{)} \\ \implies a = 1 & \text{(By definition of exponentiation)} \end{array}$$

Which would contradict a as having been defined as any element of G besides the identity. Hence m cannot equal 1, leaving us only to consider the following three other cases.

(II) $m = 3$.

In the case where $m = 3$, then it immediately follows that G contains an element of order 3 since m is the order of a by definition!

(III) $m = 9$.

Assuming now that $m = 9$, then by definition of order $m = 9$ is the smallest positive integer satisfying $a^m = 1$. So, letting $x := a^3$ (for reasons which will soon become apparent), we therefore have that:

$$\begin{array}{ll} x^3 = (a^3)^3 & \text{(By definition of } x\text{)} \\ = a^9 & \text{(By exponent laws)} \\ = 1 & \text{(By definition of order, and since } \text{ord}(a) = m = 9\text{)} \end{array}$$

Now since x and x^2 both cannot equal the identity (because if either of them did, then one of a^3 or a^6 would equal the identity, thereby contradicting our assumption that $\text{ord}(a) = m = 9$), we have that $n = 3$ is the smallest positive integer satisfying $x^3 = 1$.

Thus, by definition of order, $\text{ord}(x) = 3$, and since x belongs to G by virtue of the fact G 's a group and so must be closed under the group operation, G contains an element of order 3.

Therefore, since when $m = 3$, G contains an element of order 3, we now need to consider the last possible case.

(IV) $m = 27$.

Mimicking the approach we used for case **(III)**, assuming that $m = 27$ and letting $x := a^9$, we have that:

$$\begin{aligned} x^3 &= (a^9)^3 && \text{(By definition of } x\text{)} \\ &= a^{27} && \text{(By exponent laws)} \\ &= 1 && \text{(By definition of order, and since } \text{ord}(a) = m = 27\text{)} \end{aligned}$$

This implies that the order of x is 3 since, by the same logic argued before, $\text{ord}(x)$ being any other number would necessarily contradict our assumption that $\text{ord}(a) = m = 27$.

Thus, since $x \in G$ is an element of order 3, G contains an element of order 3 in this case as well.

Therefore, since cases **(II)**, **(III)**, and **(IV)** all imply that G has an element of order 3, and since case **(I)** cannot actually happen, G must have an element of order 3, and the statement is proven.

(b) Let G be a group of even order. Prove that G has an odd number of elements of order 2.

Solution:

To show that

(c) Let G be a group of order 12. Prove that G has an element of order 2.

Solution:

Problem 2: *The Twice-A-Prime Theorem*

Here we will prove a useful lemma about groups of order $2p$, where p is an odd prime — for example, groups of order 6, 10, 14, 22, . . .

(a) Let G be a group, and let $a, b \in G$ be two commuting elements of order 2. Prove that the set $H = \{1, a, b, ab\}$ is a subgroup of G .

(b) Let G be a group of order $2p$, where p is an odd prime. Prove that G must have an element of order p .

Solution:

(c) Combine the above steps to show that if G is an abelian group of order $2p$, where p is an odd prime, then G is cyclic.

Hint: By problems 1 (b) and 2 (b), G must have elements of order 2 and order p . Call these elements x and y . What is the order of xy ?

Solution: